



Genetic improvement of Nile tilapia (*Oreochromis niloticus*) with special reference to the work conducted by the WorldFish Center with the GIFT strain

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Abstract

We mainly (but not exclusively) draw on research and development work carried out by The WorldFish Center (WorldFish). We review in detail the current state of development of a selection program that has had a main focus on growth rate and body traits. We also present some new, unpublished, information. There is evidence of sustained gains of 10–15% per generation over more than six generations. To date, these gains have not been accompanied by any undesirable correlated response. However, the prospects of altering sexual dimorphism and the shape of the fish appear to be very limited. We also examine the issue of an appropriate environment for selection. Not surprisingly, experimental evidence on genotype by environment interactions suggests that this is more likely to be of importance when the environments in question are markedly dissimilar. We argue that no universal guidelines can be prescribed with regard to the need for more than one selection program to cope with different production environments, but rather, each case should be examined in its own right. Finally, we discuss traits likely to be candidates for inclusion in future, more elaborate, breeding objectives for Nile tilapia (*Oreochromis niloticus*), comment on selection methods that may be implemented in the future and conclude by stressing the need to maintain an effective population size in selection lines to ensure their sustainability over time.

Key words: genetic improvement, GIFT, growth rate, heritability, tilapia.

Introduction

Nile tilapia (*Oreochromis niloticus*) is a well-known tropical food fish native to Africa. To date, several selective breeding programs for Nile tilapia have been established and maintained, for example, GIFT (Genetically Improved Farmed Tilapia; Eknath *et al.* 1993; Eknath & Acosta 1998), GET-EXCEL (Tayamen 2004), FaST (Bolivar 1998) and GST (GenoMar Supreme Tilapia; Zimmermann & Natividad 2004). These selective breeding programs have generally been implemented in Asia under relatively intensive culture systems where the fish are provided with formulated feeds. Nile tilapia has not been developed significantly in Africa with a focus on the prevailing production environments in the continent, despite the fact that

Africa holds the global wealth of tilapia genetic resources and has a great natural potential for aquaculture development (Pullin 1988).

In the present review we present information on the current state of development of a selection program carried out by WorldFish that has had a main focus on growth rate and body traits (including an update on the selection response in GIFT and a study on the possibility of altering sexual dimorphism and the shape of fish), we examine the issue of an appropriate environment for selection, and we discuss traits likely to be candidates for inclusion in future, more elaborate, breeding objectives for Nile tilapia. We mainly (but not exclusively) draw on research and development work carried out by The WorldFish Center (WorldFish) with the GIFT strain.

Gupta and Acosta (2004) and Acosta and Gupta (2010) provide detailed accounts about the origin and dissemination of GIFT. In the present paper the focus is more on genetic evaluation of the strain and on research results obtained using GIFT tilapia.

Approaches to genetic improvement

Aquatic animals allow for the implementation of several approaches to genetic improvement. These include hybridization and cross breeding, chromosome manipulation, sex control, transgenesis and selective breeding. These are almost always mentioned in aquaculture genetic reports, papers and meetings without making a judgement about their relative practical value. For instance, it is seldom, if ever, stated that of all the genetic approaches only selective breeding offers the opportunity of continued genetic gain, that the gains made can be permanent, that it is the only approach in which the gain can be transmitted from generation to generation, and that gains in a nucleus can be multiplied and expressed in thousands or millions of individuals in the production sector (Ponzoni *et al.* 2007, 2008). In cases in which they are useful, the other approaches result in 'once-off' expressions of the benefit. They may be applied at the multiplication (hatchery) level, but not at the nucleus level. In the present review we focus on selective breeding.

Improvement of growth rate by selective breeding

In this section we report in some detail the results derived from the Nile tilapia selection line (Malaysian GIFT strain) jointly maintained by the Department of Fisheries (Malaysia) and WorldFish for the period 2002–2009. We present an update of the unpublished results obtained to date.

Origin of the fish

We have worked with the GIFT strain (Eknath *et al.* 1993; Bentsen *et al.* 1998; Eknath & Acosta 1998), which has been disseminated to 11 countries in Asia (Gupta & Acosta 2004). The GIFT Foundation International, Philippines, provided 63 full sib groups of 35 fish each, which were progeny from single-pair mated parents (i.e. 63 males each mated to a different female). These fish belonged to the sixth generation of selection of GIFT, and were received at Jitra, Kedah State, Malaysia, in batches towards the end of 2000 and beginning of 2001. They were mated and produced a seventh generation (base population in Malaysia) in the spawning season of 2002, which in turn produced an eighth generation in 2003. No selection took place among the fish transferred

from the GIFT Foundation because they were received in batches and there were uncertainties with regard to the environmental factors that could be influencing their performance. Two lines were created with the 2002 progeny, one selected on high breeding value for live weight (selection line), and another one selected for average breeding values (control line). With the exception of three females and three males used in 2005 and again in 2006 (Table 2 of Ponzoni *et al.* 2010), none of the parents used in one spawning season were used in a subsequent one (i.e. the generations were discrete). Note that we consider the progeny produced in the 2002 spawning season to be our base population in Malaysia, and in our analyses we treat it as part of the established control line.

Overall objectives of the project

The objectives of the project may be summarised as follows.

- 1 To maintain and continuously improve the GIFT strain of Nile tilapia and to distribute it to partner countries that are likely to benefit from its use.
- 2 To conduct research that may enhance the effectiveness of the genetic improvement program (e.g. refining the methodology with respect to management of inbreeding and effective population size, introducing new traits such as fillet yield, flesh quality and response to thermal treatment to the breeding objective).
- 3 To utilize the data collected to support capacity-building activities in the field of genetic improvement for staff from partner countries (because the GIFT population maintained at Jitra is fully pedigreed it offers great opportunities for students and staff to conduct research on the data).

Update on phenotypic and genetic parameters and response to selection in GIFT

We have recently conducted a detailed analysis of all the data collected since 2002. Tables 1–6 summarise the results. The selection response was estimated for live weight. The data included generations one to eight bred in Malaysia and were collected at Jitra Research Station. Live weights were analysed using the square-root transformation. Preliminary analyses showed that greater means were associated with greater variances in the sub classes and that the data were not normally distributed. Out of all possible transformations the square root was the one that best remedied the problem.

Tables 1 and 2 show the number of observations by year, the line (selection or control), sire and dam, and the work schedule, respectively. Numbers are such that they enable estimation of the parameters with confidence.

Table 1 Number of sires, dams and progeny by spawning season and line

Spawning season	Line	Sires	Dams	Progeny
2002	Base population	52	54	1684
2003	Selection	35	65	2560
	Control	19	19	1150
2004	Selection	54	84	3714
	Control	17	22	957
2005	Selection	42	76	1763
	Control	13	20	480
2006	Selection	49	88	3134
	Control	10	15	513
2007	Selection	41	71	4238
	Control	15	15	859
2008	Selection	52	76	2735
	Control	14	14	583
2009	Selection	51	69	2674
	Control	9	11	458
Total		473	699	27 502

Tables 3 and 4 show some simple statistics and the results of the ANOVA, respectively. The coefficients of variation of both live weight and age at harvesting were lower in more recent years, suggesting that management during the grow-out phase may have improved, thus resulting in a reduction in the variation of those two attributes. All effects fitted in the ANOVA were statistically significant, with the exception of line, which had a low probability value (approximately 0.17), but did not reach the conventional value (0.05) to deem it statistically significant. It is important to note, however, that in this analysis all generations are included and in the early generations the lines had not yet diverged. In fact, the significant spawning season by line by sex interaction can be explained by the fact that the between line difference became greater after each generation. A better estimate of genetic change is provided by the estimation of breeding values in all generations and this information is shown in Table 6. Breeding values estimate the genetic merit of individuals using not only information on their own performance, but also information from all of their relatives in the population. Sorting estimated breeding values by generation provides a way of estimating genetic change between generations.

Tables 5 and 6 show the genetic parameters and the selection response in live weight, respectively. Heritability was high, indicating that there is still abundant genetic variation and scope for further genetic improvement in growth rate. The maternal and common environmental effect (c^2) was large, most likely because of the period in which the members of full sib families are growing together in hapas until each individual is large enough to

Table 2 Schedule of reproduction and management

Spawning season	Activities				
	Mating	Nursing	Transfer to B-net	Stocking	Grow-out
2002	February–March 2002	February–April 2002	March–May 2002	April–May 2002	June–November 2002
2003	January–February 2003	January–March 2003	February–April 2003	March–April 2003	April–September 2003
2004	November 2003–February 2004	November 2003–February 2004	December 2003–March 2004	February–May 2004	March–September 2004
2005	December 2004–February 2005	December 2004–February 2005	January–March 2005	March–May 2005	March–September 2005
2006	November 2005–January 2006	December 2005–February 2006	January–March 2006	March–April 2006	March–September 2006
2007	October 2006–March 2007†	November 2006–April 2007	December 2006–May 2007	February–June 2007	February–August 2007
2008	October 2007–February 2008†	November 2007–March 2008	December 2007–June 2008	March–June 2008	March–November 2008‡
2009	January–April 2009	February–May 2009	March–June 2009	May–June 2009	May–December 2009

†The prolonged mating period was because there was not enough family for the control line and an unfavourable climate.

‡The prolonged grow-out period and harvesting period resulted from an unfavourable climate, which affected the growth of the fish.

Table 3 Number of observations (*N*), simple mean, minimum and maximum, standard deviation and coefficient of variation of live weight and age at harvesting

Variable	Spawning season	<i>N</i>	Mean	Minimum	Maximum	Standard deviation	Coefficient of variation (%)
Live weight (g)	2002	1684	227.08	29.0	608.0	101.54	45
	2003	3710	154.11	7.0	617.0	86.27	56
	2004	4671	192.87	19.0	682.0	101.38	53
	2005	2243	209.38	75.0	551.0	59.32	28
	2006	3647	223.40	49.0	532.0	69.05	31
	2007	5097	222.15	15.0	504.0	73.93	33
	2008	3318	248.82	19.7	670.0	112.58	45
	2009	3132	228.09	48.5	568.6	79.02	35
Age at harvesting (days)	2002	1684	256.65	215.0	280.0	12.77	5
	2003	3710	213.94	125.0	265.0	25.28	12
	2004	4671	244.60	173.0	302.0	28.84	12
	2005	2243	227.48	182.0	260.0	14.52	6
	2006	3647	230.83	178.0	263.0	15.97	7
	2007	5097	219.93	118.0	272.0	19.71	9
	2008	3318	273.71	214.0	311.0	21.11	8
	2009	3132	257.42	227.0	302.0	15.98	6

Table 4 ANOVA of the live weight using a square-root transformation ($LW^{0.5}$)

Effects	<i>F</i>	Probability of $>F$
Spawning season (SS)	26.75	<0.0001
Line (L)	1.91	0.1686
Sex (S)	46.03	<0.0001
Environment (E)	6.51	0.0107
SS $\times$ L $\times$ S	3.13	<0.0001
Age (SS, S, E)	180.74	<0.0001
Residual variance	2.8423	

Tests of fixed effects using PROC MIXED in SAS software (release 6.1; Cary, NC, USA).

Table 5 Variance components, heritability and maternal common environment effect for live weight using a square-root transformation ($LW^{0.5}$)

Parameter	Residual maximum likelihood estimate
Additive genetic variance (σ^2_A)	1.81
Maternal and common environmental variance ($\sigma^2_D = \sigma_{M_Ec}$)	2.05
Phenotypic variance (σ^2_P)	6.03
Heritability (standard error) [h^2 (SE)]	0.30 (0.0353)
Maternal common environment (standard error) [c^2 (SE)]	0.34 (0.0175)

be physically tagged. It would be desirable to eliminate or at least reduce the magnitude of this latter parameter.

Table 6 shows the selection response in growth rate experienced by GIFT in Malaysia. We only comment on

the response expressed in percentage terms. There was continued response over the period examined, as well as good agreement between the two methods used (52.5% comparing estimated breeding values in consecutive generations versus 52.2% comparing the selected and control lines). It is important to note that these percentages were calculated from results using the square-root transformation on the data. James (2007) shows that the percentage selection response after square-root transformation is a fraction (0.501) of that in actual units. This means that in actual units the response for the period in question was of the order of 100% (an average of 14% per generation). Coupled with the high survival consistently observed in the GIFT strain, the high potential for growth rate makes it a very attractive genetic resource. Note that the responses to selection in 2008 and 2009 are smaller than in some of the earlier generations. Under 'normal' circumstances the magnitude of response to selection varies because of sampling, and long term it may diminish owing to loss of genetic variation. In our case the lower responses in the two last years are more likely to result from adverse climatic conditions (i.e. lower than usual temperatures and greater than usual rainfall resulting in lower water quality and greater mortality) than an indication of proximity to a selection plateau.

Further evidence of genetic gain in GIFT

In a separate experiment (Khaw *et al.* 2008), genetic change in GIFT was estimated by comparing the performance of the progeny produced from cryopreserved spermatozoa from the base population originally established

Table 6 Response to selection estimated by different methods and expressed in different ways

Method	Mixed model fitted	Selection response (LW ^{0.5})†		
		Actual units (g ^{0.5})	Percentage	Genetic standard deviation units (actual/ σ_A)
Comparing the estimated breeding value for live weight between progeny from the selection line of two subsequent spawning seasons				
Between 2002 and 2003 spawning season	Fixed effects:	1.188	10.062	0.881
Between 2003 and 2004 spawning season	SS × L × S × E subclasses	1.095	9.268	0.812
Between 2004 and 2005 spawning season	Covariate:	0.986	8.348	0.731
Between 2005 and 2006 spawning season	age at harvest (SS, L, S, E)	0.733	6.206	0.543
Between 2006 and 2007 spawning season	Random effects:	0.939	7.947	0.696
Between 2007 and 2008 spawning season	animal and dam	0.584	4.947	0.433
Between 2008 and 2009 spawning season		0.672	5.691	0.498
Comparing the estimated breeding value for the live weight between progeny from the control line and the selection line of the same spawning season				
2003 spawning season	Fixed effects:	1.542	13.059	1.143
2004 spawning season	SS × L × S × E subclasses	2.691	22.788	1.995
2005 spawning season	Covariate:	3.447	29.191	2.556
2006 spawning season	age at harvest (SS, L, S, E)	4.019	34.033	2.980
2007 spawning season	Random effects:	4.955	41.957	3.674
2008 spawning season	animal and dam	5.472	46.336	4.057
2009 spawning season		6.170	52.244	4.574

SS, spawning season; L, line (selection or control); S, sex; E, environment (cage or pond rearing).

†Actual units are the differences in mean breeding values for the two methods for live weight using a square-root transformation ($LW^{0.5}$); percentage refers to actual units in relation to the least square means of $LW^{0.5}$ for the control population (11.8099 $g^{0.5}$); genetic standard deviation equals the square root of the additive genetic variance in Table 6 ($\sigma_A = 1.3488 g^{0.5}$).

Table 7 Harvest weight and survival rate least square means and standard errors for the sire generation

Sire generation	Harvest weight (g)	Survival (standard error)
Base	141 ^a (5.8)	0.86 (0.02)
Ninth	186 ^b (5.8)	0.83 (0.02)

Means with a different superscript letter are significantly different ($P < 0.05$).

in the Philippines in 1991 with that produced by freshly collected spermatozoa from the generation produced in Malaysia in 2003. The comparison involved artificial fertilization of 13 males from each generation (base and ninth) with a random sample of female brood stock. The progeny produced went through a 120-day grow-out period, after which live weight, standard length, body depth and survival were recorded. The estimated total genetic change in live weight was 64% over nine generations or 7.1% per generation (Table 7). The genetic change was lower than the estimate reported by Eknath *et al.* (1998), but in the present experiment the timespan included generations in which there was no selection. The improvement in live weight was achieved without any deterioration in survival rate, which has remained high.

After five generations of selection, Eknath *et al.* (1998) reported an annual genetic gain of 12–17%, which is considerably higher than our estimate for nine generations. Eknath *et al.* (1998) did not establish a separate control population and maintain it throughout these five generations, but rather they recreated a new control in each generation by mating a sample of average individuals for that generation. There may have been inadvertent selection of the fish to be mated as controls, and sampling problems accumulated during the selection could have caused an over estimation of the genetic gain reported by Eknath *et al.* (1998).

In contrast, the estimate of genetic gain from Khaw *et al.* (2008) may actually be biased downwards. First, frozen sperm was collected from the best growing males in the base population, whereas the sires used from the ninth generation were not the best (they were close to the average of that generation). Second, the formal GIFT selection program ended at the fifth generation, and there was no selection when mating was conducted to produce the sixth and seventh generations. The sixth generation was sent to Malaysia and was used there to establish the population now located in Jitra Aquaculture Extension Center, Kedah state. That means that in between these nine generations, there are two in

which no selection of superior individuals for growth rate or any other trait was conducted. If allowances were made for this when calculating the genetic gain per generation, a figure (9%) closer to that reported by Eknath *et al.* (1998) would be obtained. Note also that the genetic change reported in the previous section of this paper based on the GIFT population established in Malaysia is in close agreement with that reported by Eknath *et al.* (1998).

In Africa, where very little has been done in terms of genetic improvement of Nile tilapia, one may safely assume that productivity of the current stock is at the level of the GIFT base population or lower (Brummet *et al.* 2004). Hence, one may also safely assume that the introduction of GIFT to Africa would improve growth by at least 64%. This is not a trivial gain and could greatly benefit emerging aquaculture industries in many African countries.

Environment for selection

Traditionally, tilapia has been cultured in earthen ponds under extensive and semi-intensive systems (El-Sayed 2006). Owing to a shortage of freshwater and the widespread distribution of tilapia culture, farmers and commercial entrepreneurs have in many cases shifted to more intensive systems in ponds, tanks and cages (El-Sayed 2006). With the wide range of culture systems and environments between countries and within a country (particularly for large countries, such as China, or whole regions such as Sub Saharan Africa) ascertaining the appropriate environment for selection is important. Furthermore, over the past two decades, improved Nile tilapia strains, for example, GIFT (Eknath *et al.* 1993; Eknath & Acosta 1998), GET-EXCEL (Tayamen 2004) and FaST (Bolivar 1998), have been widely distributed from Asia (the continent where those fish were selected) to other parts of the world. These two features of present tilapia production clearly indicate the need for genotype by environment interaction ($G \times E$) studies in Nile tilapia breeding programs.

Several $G \times E$ studies have been carried out in different countries. In Malaysia, the growth performance of GIFT was investigated in two main culture environments, earthen ponds and cages. This study (RW Ponzoni, NH Nguyen, HL Khaw, A Hamzah, KR Abu-Bakar and HY Yee, 2010, unpubl. data) shows that the genetic correlation for harvest weight between pond and cage was 0.70 ± 0.11 . In another $G \times E$ study with GIFT conducted in the Philippines seven different environments encompassing a range of farming systems and agro-climatic regions were examined (Eknath *et al.* 2007). Live weight expressions in all environments were positively correlated

with each other, and the correlations ranged from 0.36 to 0.99. The correlations were generally high within the pond environments (0.76–0.99) and within the cage environments (0.99), but lower in some cases and more variable between the pond and cage environments (0.36–0.82). These results illustrate the notion that the greater the difference between environments, the greater the chance of finding an important $G \times E$.

This latter concept is supported by a recent $G \times E$ study with GIFT by Luan *et al.* (2008) in Vietnam. They studied harvest weight and survival of GIFT growing out in brackish and in freshwater systems. The genetic correlations between these two environments are 0.45 ± 0.09 and 0.42 ± 0.05 for harvest weight and survival, respectively. The authors conclude that given these correlation values two separate selection programs should be undertaken for the improvement of these traits in brackish and in freshwater environments.

Khaw *et al.* (2009) carried out a $G \times E$ study with Nile tilapia at the Regional Research Station of The WorldFish Center at Abbassa, Egypt. From the same brood stock, two selection lines were developed: one in which all progeny were reared in a low input environment (only received chicken manure as a daily external nutrient source) and another one maintained in a high input environment (received formulated pellets that contained 25% protein). The dataset consisted of 7640 animals (progeny of 298 sires and 493 dams) with body weight records at harvest over three discrete generations. With regard to the statistical analyses to examine the $G \times E$, four different models (with different fixed effects) were fitted in a bivariate animal mixed model to estimate the relevant genetic parameters (harvest weight in each environment was treated as a different trait). The genetic correlation for body weight between the low and high input environments ranged from 0.74 to 0.84 (with a standard error of 0.15–0.36). Unfortunately, Khaw *et al.* (2009) found some degree of confounding between the genetic and environmental effects. This was mainly the result of limitations in the data structure (discrete generations and without a control line), which did not enable a neat separation of genetic and environmental effects from each other.

Robertson (1959) suggested that when the genetic correlation between the trait expressions in two different environments was <0.8 the $G \times E$ could be considered unimportant. However, no universal guidelines for the handling of $G \times E$ in animal breeding have been developed, and owing to the nature of the problem, quite likely, never will. Thus, we should treat every case based on its nature, merits and economic aspects (James 2009). For example, in the study by Khaw *et al.* (2009) that was carried out in Egypt, the lower limit for the genetic correlation for harvest weight in both environments was

less (0.74) than 0.8, which following Robertson's (1959) suggestion would indicate that two different selection programs would be required. Note, however, that, for instance, if the selection program were conducted in the low input environment, the genetic correlation indicates that 74% of the gain would be captured in the high input environment. Owing to the fact that Egypt is a developing country with limited resources, in this situation it would be difficult to justify having two breeding programs for one species. In such circumstances, decisions should be made by examining the statistical evidence on $G \times E$, the likely future evolution of the environmental conditions and the economic aspects of the country in question in relation to the resources available for genetic improvement programs (Montaldo 2001).

In the future, with the rapid expansion of the Nile tilapia aquaculture industry, more in-depth $G \times E$ studies should be carried out by focusing on growth performance and survival rate in a range of different realistic culture environments (e.g. ponds, cages and tanks), culture systems (e.g. extensive, semi-intensive and intensive) and agro-climatic environments (e.g. brackish and freshwater, tropical, subtropical and temperate, high and low seasonality) likely to prevail in the foreseeable future.

The issue of $G \times E$ interactions may be looked at from a less conventional viewpoint. In general, when genetic improvement programs are established in developing countries a single genetically improved line is developed for each species of interest. This poses the obvious problem that that single line is required to service a range of production environments. The solution calls for a breeding strategy that goes beyond common animal breeding practice. The term phenotypic 'plasticity' is used more often in aquatic animal literature than in livestock. Phenotypic plasticity refers to organisms of the same genotype developing systematically different phenotypes in different environments. Genotypes showing highly variable phenotypes across environments are 'plastic', whereas genotypes that show little variability are 'robust' (De Jong & Bijma 2002). The concept of robustness has been advocated as a breeding goal (Ellen *et al.* 2008) because it would have a positive effect on welfare and health without loss of integrity. Robust animals are organisms that have the ability to maintain their performance across different environmental conditions (Waddington 1960). Selection for robustness in fish breeding programs would result in animals that have a greater ability to maintain their homeostatic state when exposed to stressors, thus improving fish welfare (Ellen *et al.* 2008). James (2009) discusses this issue in the context of $G \times E$ interactions. When a genetic improvement program has to service several environments, selection index theory could be

applied using the relative importance of the environments as weighting factors. Because the breeding goal should target the future circumstances the weighting factors should reflect the anticipated future relative importance, rather than the present one. Given the high reproductive rate of aquatic animals and the relatively modest resources (compared with livestock) that would be required for testing in multiple environments it appears that research in this area would be worthwhile. To our knowledge there have been no genetic improvement programs with aquatic animals formally targeting multiple environments.

Production of all male populations for grow-out and marketing

Producing 'all male' tilapia populations for the production system

Monosex culture of male tilapia is often preferred to a mixed sex system because of sexual dimorphism, that is, males are substantially larger than females (e.g. Ponzoni *et al.* 2005; Nguyen *et al.* 2007). In addition, precocious reproduction of this species also leads to low growth performance of stocked fish as a consequence of overcrowding and feed competition in production ponds. There are several methods available for the creation of monosex tilapia populations for the production system, such as manual sexing, interspecific hybridization, androgenesis, triploidy, transgenesis, hormonal sex reversal and YY male technology (Beardmore *et al.* 2001). However, the efficiency of these technologies varies with populations, seasons of fry production and culture environments. To date, all available 'all male' production methods have suffered from one or more limitations that make them either not cost-effective, unsustainable or not acceptable. For instance, masculinizing tilapia with hormones has raised consumers' concerns regarding food safety. In the case of YY technology, the creation of YY males takes three generations of breeding. This means that if the source population is undergoing selection, by the time the YY males are ready for use they will be lagging three generations behind in terms of genetic gain. In practical terms this could be between 20% and 45%, depending on the rate of progress in the selected population. Furthermore, the application of YY technology relies on the participation of a laboratory with relatively advanced facilities for the production of YY males. It creates a relationship of dependence of the hatchery producing the all male progeny for the production system on the laboratory providing the YY males. This is not a scenario to be favoured in general, and even less so in developing countries. Other techniques also have limitations, but we do not discuss them in the present review. Hence, alternative means of

producing all male tilapia populations merit further research (e.g. increasing the male to female ratio in response to thermal treatment).

A novel method: thermal treatment

Production of monosex tilapia (males) can be achieved with a high rate of success by temperature treatment (e.g. Baroiller *et al.* 1995; Baroiller & D'Cotta 2001). Across studies, a high temperature ranging between 36 and 38°C applied over 10 days post-hatching (the critical sensitive period for sex differentiation) can induce a high proportion of males (above 80%). A number of studies have suggested a possible temperature by genotype interaction (Baroiller *et al.* 1995; Abucay *et al.* 1999; Baroiller & D'Cotta 2001; Tessema *et al.* 2006). There is also large between family (within population) variation in the level of thermo-sensitivity to treatment, as recently demonstrated by Tessema *et al.* (2006) and Bezault *et al.* (2007). In summary, there is potential for masculinization of female genotypes by temperature treatment. However, high temperatures can also have both masculinizing and feminizing effects depending on the sex genotype, origin of the fish and various genetic and environmental factors affecting the sex determination mechanisms.

At The WorldFish Center, we have conducted preliminary studies to investigate the effect of heat treatment on sex ratio, growth and survival in the GIFT strain (RW Ponzoni, NH Nguyen, HL Khaw, A Hamzah, KR Abu-Bakar and HY Yee, 2010, unpubl. data). Experiments with various levels of temperature applied at different rearing periods have been conducted under both laboratory and field conditions. In each experiment, newly hatched fry of each family (soon after egg yolk absorption) were split into two or three groups of equal size. One group was reared under a normal water temperature (between 28 and 30°C; control) and temperature treatments were applied to the other groups, that is, the water temperature was set at 36°C or at 38°C and kept constant by special heaters for 10 and 6 days, respectively. After the treatment period, the fry of each family were returned to normal water conditions in separate hapas installed in tanks, or transferred to nursery hapas in ponds. All groups were subject to the same feeding and management regimes throughout the experimental period. Under the laboratory test, fingerlings of an average size of 4 g were sexed using the aceto-carmin staining method. Manual sexing was conducted for the fish under field conditions. At sexing, measurements of body weight and standard length were recorded for individual fish. Across the experiments, there was a greater percentage of males in the treatment groups than in the control (63.3% versus

59.0%), although a chi-squared test showed that the differences between the treatment and control groups were not statistically significant. There was variation in the sex ratio among families, ranging from 20% to 80% of males in some families. There were no statistical differences in body traits between the control and heat treatment groups. Survival of fry from hatching to sexing was in general lower (4–10%) in the heat treatment group than in the control group. A new series of replicate experiments is now underway, the principal aim of which is to define a suitable experimental protocol to increase the male to female ratio in the GIFT strain by thermal treatment. If the GIFT strain responded to the heat treatment as reported in other studies, it might be possible to conduct selective breeding for sensitivity to the treatment. In a recent study conducted by Wessels and Hörstgen-Schwark (2007), two generations of selection resulted in a male percentage of 90%, with a realized heritability of 0.69.

In addition to conventional selective breeding to increase the male to female ratio, in the future, if novel genes with large effects are successfully detected, a combination of marker assisted selection and polygenic selection could significantly increase the rate of response for thermo-sensitivity. The outcome of such a study could help provide a practical and totally acceptable technology for sex control in tilapia and perhaps for other fish as well. Genetically divergent selection lines developed from conventional genetic selection could be used for subsequent studies on physiological and molecular genetics mechanisms of sex determination in this strain. Some studies have characterized genes, for example MM20C which is differentially expressed in temperature-masculinized females (D'Cotta *et al.* 2001), or CYP19 genes which encode aromatase cytochrome P450 enzyme. The P450 enzyme catalyzes the formation of oestrogens from androgens. A suppression of aromatase gene expression can result in masculinization associated with functional testis development in tilapia. However, the regulation of aromatase genes is presently not well understood. Thus, one important component of a study in the area of genomics would be to focus on the identification of quantitative trait loci (QTL) and candidate or functional genes associated with sex ratio in GIFT. The successful detection of major genes would increase the rate of genetic progress through a combination of marker-assisted selection (MAS) with conventional selective breeding.

Finally, success in improving growth rate by selective breeding has somewhat reduced the importance and perceived need to have all male tilapia populations in the production system, unless the aim is to market the fish at live weights <800 g. The market size of individuals in

most tilapia production countries normally ranges from 200 to 600 g. Within this range, performance of GIFT is little affected by sexual maturation because selection has markedly increased the growth rate of fish (e.g. achieving 600 g within 3 months of culture under a standard farming environment in Malaysia; A. Hamzah, 2010, pers. comm.). From a management point of view, minimizing variation in body traits can be done through improved management practices, such as synchronization of fry production, improved nutrition and reduced stocking density.

Fillet yield and flesh quality

Consumers and the aquaculture industry are increasingly focusing their attention on flesh and eating quality. In some countries, the marketing of fillets instead of whole fish has become common for both domestic and export markets. One would be justified in thinking that in the not too distant future the marketing system could in some cases shift from payment based on whole fish live weight to fillet weight, in a manner similar to what has happened with livestock (McLeod 2003). Thus, there is a need to examine the possibility of broadening the breeding objective in GIFT and other strains to include yield and flesh quality as additional traits.

Strain comparisons

Red tilapia is the preferred strain in Malaysia with a reputation of having very good flesh quality. Khaw *et al.* (2006) and Ponzoni *et al.* (2006) carried out a series of sensory evaluation studies on the flesh quality and fillet traits of GIFT and red tilapia. Regarding the flesh-quality studies, the results show that although there were significant differences in some sensory attributes between GIFT and red tilapia, these were not consistent (differed between tests, but did not consistently favour one of the strains). However, the more important finding was that the score for all of the sensory attributes for both strains fell within the range of highly acceptable flesh. The analysis of fillet traits showed a similar pattern in all of the experiments conducted (Khaw *et al.* 2006; Ponzoni *et al.* 2006). GIFT emerged as a valuable strain for filleting, but not because of an advantage in fillet yield (which was very similar between both strains), but rather because of its greater fillet weight owing to its greater growth rate. The fillet yields reported in the studies by Khaw *et al.* (2006) and Ponzoni *et al.* (2006) ranged from 26.66% to 38.89%, and are in agreement with the values reported by Rutten *et al.* (2004), which are in the range of 26–37%. From their studies on flesh quality and fillet traits, Khaw *et al.* (2006) and Ponzoni *et al.* (2006) concluded that

GIFT is a high-performing strain in terms of growth rate and fillet traits. Although no important differences were found in the sensory evaluation, it can be concluded that GIFT is comparable to red tilapia (the preferred strain in Malaysia) in that respect, hence totally acceptable to Malaysian consumers, which makes it a strain of great potential for both domestic and export markets.

Within strain estimation of genetic parameters and selection responses

We conducted a study examining four main groups of traits: (i) fillet weight and yield; (ii) chemical composition; (iii) flesh quality attributes; and (iv) fatty acid composition. Data were collected over three generations from the long-term selection program in the GIFT strain of Nile tilapia in Malaysia. During 2006, 2007 and 2008 data on fillet weight and yield were recorded in 5332 fish from 174 sires and 280 dams. Fillets were weighed and frozen for later use in the assessment of flesh quality attributes. Approximately 2000 fillets were randomly sampled across families, selection lines, sex and batch of filleting within generations, and were sent to a specialized laboratory for an analysis of flesh quality attributes, including protein %, moisture %, fat %, pH and colour. A subset of the samples (514 samples of offspring from 104 sires and 154 dams from two generations 2006 and 2007) was also analysed for fatty acid composition. We assembled a full dataset including body measurements (live weight, standard length, body width and body depth) and fillet traits (fillet weight and fillet yield), chemical composition (protein, fat, moisture content), two important parameters of flesh quality attributes (pH and colour) and fatty acid composition (22 identified fatty acids and their derivatives) for genetic analysis. A multi-trait model using the restricted maximum likelihood method was applied to a full pedigree comprising a total of 18 970 animals to estimate genetic parameters and selection responses. The heritability was calculated as the ratio of additive genetic to total phenotypic variance. In a similar manner, common environmental and maternal effects (resulting from separate rearing of full-sib groups before tagging in addition to the additive genetics) were also estimated. The correlations between two traits were computed as the covariance divided by the square root of the product of the variances of the two traits in question. Genetic changes were measured as the difference in least square means between the selection and control lines.

Heritabilities

The estimates of heritability for body and carcass traits were of moderate magnitude (0.20–0.33). The

proportions of variance explained by the maternal and common environmental effects (c^2) were small, ranging from 4% to 8% for body traits and fillet weight (Nguyen *et al.* 2010a). For proximate composition and flesh quality attributes, the heritability estimate for protein content was low, whereas the heritabilities for fat and moisture contents as well as for flesh parameters (pH and colour) were generally moderate (Nguyen *et al.* 2010b). The c^2 values were not significant for carcass and quality traits. In GIFT tilapia, the heritabilities for fatty acids varied from zero to medium (Nguyen *et al.* 2010c). These results indicate possibilities for genetic improvement of flesh quality traits in the GIFT strain through conventional selective breeding. However, different selection strategies would be required because the magnitude of genetic variation differed between traits, particularly between fatty acid groups (saturated versus monounsaturated or polyunsaturated).

Genetic correlations

The genetic correlations between fillet weight and body measurements were very high (0.78–0.96), indicating that both growth and carcass performance can be simultaneously improved (Nguyen *et al.* 2010a). Fillet weight showed greater correlations with body weight than with other body size traits (e.g. length). The genetic correlations between fillet yield and body traits were generally low (0.35–0.44), suggesting that harvest weight is the most effective selection criterion to improve both growth and fillet weight in practical breeding programs. There was also a high, positive genetic correlation between fillet weight and fillet yield ($r_g = 0.68$). Breeding objectives for increased fillet weight are thus expected to result in improvement in fillet yield. This approach is practically feasible for two main reasons: (i) the accuracy of prediction equations for fillet weight based on body measurements was very high, $R^2 = 0.94$ (Nguyen *et al.* 2008); and (ii) non-requirement for slaughter of live animals. Regarding the genetic correlations between body traits and flesh composition, the estimate between protein content and body weight was very low. The genetic correlations between fat and body weight were moderate to high and mostly positive. The estimate between moisture and body weight was positive in GIFT tilapia. Flesh colour exhibited a moderate to high positive correlation with body weight (Nguyen *et al.* 2010b).

Nguyen *et al.* (2010c) examined genetic relationships between fatty acids and body, carcass, proximate chemical composition and flesh quality traits. Among the fatty acids, the genetic correlations between fatty acid groups were generally antagonistic. In contrast, the genetic correlations of fatty acids within the same group were mostly positive. The genetic associations between

important high chain polyunsaturated fatty acids and performance, fillet and flesh quality traits were variable, namely, with estimates both favourable and unfavourable.

Selection responses

The effects of selection for increased performance on flesh quality were also examined in a series of experiments in GIFT tilapia. To the best of our knowledge, there have been no comparable studies in other aquaculture species.

Carcass traits. The selection program for high growth rate in GIFT has resulted in a significant increase in fillet weight. In our study, fish were filleted at an average body weight of 527 g and fillet yield averaged 33.6%. The accumulated response in fillet weight up to the latest generation of selection included in the carcass study (corresponding to the spawning season in 2008) was 23% (Nguyen *et al.* 2010a). In contrast to fillet weight, change in fillet yield was not significant. The lack of change in fillet yield was consistent with the genetic gain in weight at filleting, which was 24%. Both body weight and fillet weight changed at the same rate, hence fillet yield, being a ratio between these two traits, remained unchanged. Our results in tilapia were consistent with those reported in livestock species, indicating that the improvement in growth in the GIFT strain has not been the result of an increase in some worthless body part, but rather in a very valuable part, the fillet. This finding is very important in assessing the consequences of the genetic improvement program, and of course for the fillet market.

Flesh composition and flesh quality attributes. We also examined fillet composition (protein, fat, moisture content) and two important flesh quality parameters (pH and colour) in the GIFT strain. Mixed model analyses showed that flesh composition traits in the selection line did not differ from the control.

Fatty acid composition

In total, 22 fatty acids were identified in the GIFT fillet. Our results show that there were no major changes in fatty acid composition as a consequence of long-term selection for high growth in the GIFT strain (Nguyen *et al.* 2010c). This is partially explained by the non-significant difference in fillet fat content between the selection and control lines. The negligible changes in fatty acid composition of GIFT indicate that selection for high growth has had a very limited impact on fatty acid composition.

In summary, our results in GIFT tilapia have indicated that there is genetic variation in flesh quality traits, and

this provides scope for genetic improvement. However, the inclusion of these traits, particularly fatty acids, in breeding objectives may be fraught with difficulties for a number of reasons: (i) high cost of chemical analysis; (ii) lack of efficient methods of data recording, for example, measurements of flesh quality require the slaughter of live animals; (iii) complex biological and physiological control; and (iv) lack of pricing systems that reward the producer. The benefit of including quality traits in breeding programs depends greatly on their economic values. Unless pricing systems place a high emphasis on meat quality and the cost of routine data collection (i.e. after slaughter on relatives of selection candidates) is reduced, conventional selection may not be justified. Currently tilapias are priced on the basis of their wet body weight, and the market could in the future shift from payment based on whole fish live weight to fillet weight, but it is less likely that it will be based on flesh quality or fatty acid composition. Hence, fillet weight, a carcass trait of paramount importance, should be primarily targeted for improvement in any aquaculture species. Fillet weight can be effectively improved through indirect selection for growth related traits as evidenced in the GIFT strain in our study. The breeding objectives targeting the improvement of fillet weight are expected to result in a correlated increase in fillet yield, although the magnitude of the response is predicted to be relatively small (Nguyen *et al.* 2010a). The expansion of the breeding goal in GIFT to include other flesh quality traits, particularly desirable fatty acids, would be justified only when the market moves towards a payment system based on flesh quality. This is simply because the breeding objective should include traits that impact on the income and expenses of the farmer.

The set of genetic parameters obtained in this study enables the prediction of possible changes that may result from long-term selection for growth performance. They can also be used for modelling alternative selection strategies to improve the meat and eating quality of the GIFT strain. Most importantly, we found from a series of studies in flesh quality that there was very limited impact of selection for increased growth rate on flesh quality traits. This is a very significant finding because most breeding programs for aquaculture species have almost exclusively focused on improvements in body weight. However, a close monitor of the correlated changes that may occur as a result of selection for high productivity is recommended in all genetic improvement programs.

A future selection direction?

There are issues related to animal behaviour and animal welfare that need to be examined when using newly

developed (and developing) quantitative genetic theory. These issues constitute a fertile ground for research and development with aquatic animals. With the rapid expansion of the aquaculture industry (FAO 2009), we can foresee that fish welfare will become one of the major challenges in fish breeding programs. Welfare can be linked to product quality and quantity and to production efficiency (Ashley 2007). Competition among individuals may be one of the causes of the frequently observed high coefficient of variation for harvest weight (e.g. Ponzoni *et al.* 2005). There is no universal definition of fish welfare, but it is commonly defined as representing the physical and mental state of well-being of the animal in relation to its internal and external environment (Ellis *et al.* 2002). Farmed fish are often subjected to stress during crowding, handling, transport and controlled reproduction (Kubilay & Uluköy 2002). In developing countries, fish selection programs have been mainly focused on growth rate. Such a breeding objective favours the selection of more aggressive animals (Lopez 1996). Social interactions among individuals can have profound influences on the expression of performance and welfare traits (Muir 1996, 2005; Brichette *et al.* 2001; Denison *et al.* 2003; Bjima *et al.* 2007). They can reduce growth because of competition for limited resources and result in mortality as a result of cannibalism. The stress effects from high stocking density under intensive culture systems (Ellis *et al.* 2002) coupled with such behavioural change may act against the welfare of genetically improved farmed fish. A strong case can be made for selective breeding programs to include behavioural traits that may lead to reduced aggression, greater uniformity in harvest weight and that are related to fish health and welfare. We have recently started experimental work in this area with GIFT, in collaboration with Wageningen University.

Conclusion

The WorldFish Center has played a pioneering role in the initiation and conduct of genetic improvement for aquatic animal species in developing countries. This leading role should in many cases continue. We approach work in this area in a logical and systematic manner, by addressing, as deemed appropriate in each circumstance, all the activities that the planning, design and conduct of a genetic improvement program entail.

- 1 Description or development of the production system(s),
- 2 Choice of the species, strains and breeding system,
- 3 Formulation of the breeding objective,
- 4 Development of the selection criteria,
- 5 Design of the system of genetic evaluation,
- 6 Selection of the animals and of the mating system,

- 7 Design of the system for expansion and dissemination of the improved stock, and
- 8 Monitoring and comparison of alternative programs.

This approach is not only useful in itself in the sense that it enables a logical treatment of the subject matter, but it is also helpful in the identification of areas in which knowledge or its application are deficient, and that should therefore become the target of research, development and technology transfer. As genetic improvement programs unfold, limitations and areas where there is room for refinements are identified. These, then become focused research areas, the results of which feed back into the program making it gradually more effective.

At present, particularly in newly initiated genetic improvement programs, growth is likely to continue to be the main focus of selection. This is justified because of the great economic importance of the trait in the production system given the very high reproductive rate of fish. Most developing countries completely lack improved strains so initially, despite possible evidence about $G \times E$, a single program will have to service a range of production environments. Later, as the program progresses, becomes consolidated, and production systems get better defined, resources permitting, specialized programs servicing specific production systems may be developed. With the development of more sophisticated markets (i.e. demanding product quality specifications), either domestic or export, the need to select for fillet yield and flesh quality traits may emerge. This will add complexity to the programs and inevitably, fully pedigreed populations and BLUP (Best Linear Unbiased Prediction) evaluations for the estimation of genetic merit of such traits will have to be conducted. Selection for less conventional traits, such as sensitivity to thermal treatments with the purpose of producing all male progeny, may have to be integrated as well to increase the efficiency of some production systems. Selection for disease resistance has not emerged as a priority in tilapia, but it may in the future as production systems intensify. The classic selective breeding technology based on quantitative genetics can serve selection programs for growth rate and survival well, but for the more difficult to measure traits, such as carcass and flesh quality, sensitivity to thermal treatment and disease resistance, molecular techniques would be very valuable if they enable selection on markers or directly on the genes affecting the trait. Our perception is that in the foreseeable future simple programs totally based on quantitative genetics will co-exist with much more sophisticated ones using quantitative genetics jointly with state-of-the-art molecular genetics techniques, each program effectively servicing a specific set of production, economic and social conditions.

Finally, it is important to note that reporting about improved strains usually focuses on their current superiority with regard to production traits and on the advantage they would confer to production systems using them. However, in addition to such immediate goals we should think about the long-term sustainability of that superiority, which can be quantified by studying the build-up of inbreeding and the resulting effective population size in the strain in question. Ponzoni *et al.* (2010) thoroughly examined this issue in the Malaysian GIFT strain. Their conclusions are that from a broad genetic perspective, the effective population size of GIFT has to be maintained at a level that permits: (i) sustainable genetic improvement and containment of inbreeding; and (ii) adaptation of the population to changing circumstances, such as changes in environmental conditions or expanded breeding objectives (Notter 1999; Bijma *et al.* 2002). With regard to the first point, in the case of GIFT in Malaysia, we have achieved an effective population number that is above a critical level that would put the sustainability of the program in the short term (10 years) future at risk (although there is not much room for complacency with an effective population number of the order of 88). In contrast, in relation to the second point, our population falls far below the minimum recommended of 500, and we cannot entertain the illusion that it will be able to adapt to any substantially different future requirements.

The number of families we produce from GIFT each generation and the number of progeny we rear are dictated by the size of the facilities we have access to, not by statistical considerations. If we had had access to more resources, we would have aimed for the production of more families per year. It is extremely difficult to get the message through to administrators and to donors about the need for resources to conduct sound genetic work, despite the undisputed economic benefits that can be derived from tilapia and carp genetic improvement programs (Ponzoni *et al.* 2007, 2008).

The rapid transfer of genetic improvement programs from institutions or from government agencies to the private sector is often advocated as advantageous. The promoters of such an approach ignore the reality that the private sector will be concerned primarily (if not exclusively) with immediate economic benefits, and not at all with long-term considerations about retaining evolutionary potential. Our findings, coupled with those of Ponzoni *et al.* (2007, 2008) on the economic benefits of genetic improvement, support the notion of involving agencies concerned with the long-term future of aquaculture, and not only on immediate benefits. Vertically integrated enterprises appear to be a good alternative if they have a long-term commitment to the industry.

We know of no studies similar to the present one among aquatic animal species. Given that the application of quantitative genetics in these animals is relatively recent, and that often a single selection line of a particular species is kept in a country or region, we believe that additional studies, similar to ours, in other selection lines would be of value.

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